Md Shakhrul Iman Siam

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Research Interest

My research centers on **multimodal LLMs**: building multimodal foundation models and agentic systems that perceive long-horizon, real-world signals from egocentric video, gaze, and audio, and reason over them efficiently.
Keywords: LLM Agent, Egocentric Long Video Understanding, Long-Context and Memory, Agentic Workflows, Audio-Language Models, Automatic Speech Recognition (ASR), Reinforcement Learning (RL).

Work Experience

Microsoft Redmond, WA, USA
- Applied Science Intern May 2026 - August 2026

Research Focus: LLM Harness, Agentic workflow

- **LLM Agent for PowerPoint Slide Generation**
  - Built task-complexity-based model routing for PowerPoint Agent, reducing inference COGS by routing simple tasks to cheaper models with frontier LLM fallback.
  - Replaced VLM-based visual QA with deterministic scripts for overlap, off-slide-edge detection, and layout operations.
  - Reduced 21% average tokens and 16% latency by eliminating redundant visual passes without affecting performance.

Microsoft Remote, USA
- Research Intern June 2025 - August 2025

Research Focus: ASR, Voice Privacy, Neural Audio Codec

- **Neural Audio Codec for Secured Voice Transmission [IWAENC 2026]**
  - Designed a Neural Audio Codec that disentangles speaker identity from voice data and transmits with a very low bitrate.
  - Achieved 13.5% improvement of Equal Error Rate compared to state-of-the-art Voice conversion system with only 6.6% degradation of relative WER compared to original audio.
  - Fine-tune an LLM-based Automatic Speech Recognition System that uses the transmitted codes directly as input without the need to reconstruct to the audio waveform.

The Ohio State University Columbus, Ohio, USA
- Graduate Research Associate, OSU AIoT and Machine Learning Systems Lab August 2023 - Present

Advisor: Prof. Mi Zhang

- **Memory Construction and Agentic Retrieval for Long-Horizon Egocentric Video Understanding**
  - Designed a structured memory system integrating visual observations, speech transcripts, and episodic summaries for long-horizon video question answering.
  - Developed an agent harness with eight retrieval tools for iterative evidence gathering and multi-step reasoning.
  - Trained the agent using reinforcement learning (RL) with GRPO to learn adaptive tool selection and retrieval strategies.
- **Gaze-guided Keyframe Selection for Egocentric Long Video Understanding**
  - Developed a training-free keyframe selection framework for long egocentric video understanding, achieving up to 15.7% accuracy improvement over SOTA sampling methods across various Vision Language Models(VLMs).
  - Designed a cross-modal query-frame retrieval pipeline and Maximal Marginal Relevance (MMR) for keyframe selection.
- **Multimodal Audio-language Foundation Model for Respiratory Health Diagnosis [ACL 2026]**
  - Developed a Multimodal Foundation model that will capture sounds from a smartphone microphone and perform several downstream tasks like disease (Covid-19, TB, COPD) detection, generating health reports, and Question-answering.
  - Utilize LLMs with PEFT and LoRA to fine-tune multimodal models from cough sounds and patient text metadata.
- **A large-scale multi-modal reading dataset of 100 hours of egocentric videos [NeurIPS 2025]**
  - A first-of-its-kind dataset that enables egocentric and spatial AI on the next-generation Meta AI glasses.
  - Developed and trained transformer-based model to detect reading activities using different input modalities (e.g., egocentric video, eye gaze, IMU)

ACI Limited Dhaka, Bangladesh
- Machine Learning Engineer October 2021 - July 2023

Domain Experience: Computer Vision, Speech Recognition, Predictive analysis, NLP

- Developed a system to re-identify and track customers' movement in Retail outlets from CCTV cameras.
- Implemented face detection using Retinaface and face recognition using Arcface. Converted the model to TFLite to improve inference speed by 4×.
- Medicine name recognition system from handwritten prescription. Developed OCR models using EasyOCR, PyTesseract, and PaddleOCR to recognize medicine names.
- Implemented a voice-controlled conversational chatbot by integrating a streaming ASR model and a Text-to-speech model for automated speech recognition.

Education

- **The Ohio State University** — Columbus, Ohio, USA
  Ph.D. in Computer Science and Engineering (CSE) — 08/2023 - 08/2027 (Expected)
- **The Ohio State University** — Columbus, Ohio, USA
  M.S. in Computer Science and Engineering (CSE) — 08/2023 - 12/2026
- **Bangladesh University of Engineering and Technology (BUET)** — Dhaka, Bangladesh
  BSc. in Electrical and Electronic Engineering (EEE) — 02/2016 - 02/2021

Publications

Computer Vision

1. [**Mobisys 2026**] MemoLens: Empowering Augmented Reality Glasses with Super Memory. Samiul Alam, *Shakhrul Iman Siam*, Mi Zhang
2. [**NeurIPS 2025**] Reading Recognition in the Wild. Charig Yang, Samiul Alam, *Shakhrul Iman Siam*, Michael J. Proulx, Lambert Mathias, Kiran Somasundaram, James Fort, Omkar Parkhi, Yuheng Ren, Mi Zhang, Yuning Chai, Richard Newcombe, Hyo Jin Kim
3. [**Under Review**] SuperMemory-VQA: An Egocentric Visual Question Answering Benchmark for Long-Horizon Memory. Samiul Alam, *Shakhrul Iman Siam*, Michael J. Proulx, James Fort, Richard Newcombe, Hyo Jin Kim, Mi Zhang
4. [**Under Review**] GazeFrame: Gaze-guided Keyframe Selection for Egocentric Long Video Understanding. *Shakhrul Iman Siam*, Samiul Alam, Mi Zhang
5. [**ICECE 2022**] A Deep Learning-Based Person Detection and Heatmap Generation Technique with a Multi-Camera System. *Md Shakhrul Iman Siam*, Subrata Biswas
6. [**SPICSCON 2019**] A Dual-Purpose Refreshable Braille Display Based on Real-Time Object Detection and Optical Character Recognition. KM Naimul Hassan, Subrata Kumar Biswas, Md Shakil Anwar, *Md Shakhrul Iman Siam*, Celia Shahnaz

Audio, Speech and NLP

7. [**ACL 2026 (Main)**] RespiraMFM: A Multimodal Foundation Model with Contrastive Audio-Language Alignment for Respiratory Disease Identification. *Shakhrul Iman Siam*, Tiantian Feng, Jiankun Zhang, Shrikanth Narayanan, Mi Zhang
8. [**EMNLP 2026 (Main)**] ChildVox: A Speech, Audio, and Large Audio-Language Model Benchmark in Understanding and Characterizing Sound across Childhood. Tiantian Feng, Anfeng Xu, Xuan Shi, Aditya Kommineni, *Shakhrul Iman Siam*, Megan Micheletti, Zhonghao Shi, Helen Tager-Flusberg, Mi Zhang
9. [**IWAENC 2026**] DECAF: A Privacy Preserving Speech Codec Using Speaker Disentanglement and Canonical Voice Conversion. *Shakhrul Iman Siam*, Dushyant Sharma, Stanislav Yu. Kruchinin, Peter Skala

Signal Processing

10. [**TOSN 2024**] Artificial Intelligence of Things: A Survey. *Shakhrul Iman Siam*, Hyunho Ahn, Li Liu, Samiul Alam, Hui Shen, Zhichao Cao, Ness Shroff, Bhaskar Krishnamachari, Mani Srivastava, and Mi Zhang.
11. [**Mobisys 2024**] ChirpTransformer: Versatile LoRa Encoding for Low-power Wide-area IoT. Chenning Li, Yidong Ren, Shuai Tong, *Shakhrul Iman Siam*, Mi Zhang, Jiliang Wang, Yunhao Liu, and Zhichao Cao.

Skills Summary

Programming	Python, C/C++, CUDA, Bash/Shell, SQL, MATLAB
ML Frameworks	PyTorch, HuggingFace (Transformers, Datasets, Accelerate), PEFT/LoRA, TRL, DeepSpeed, FSDP, scikit-learn, NumPy, Pandas
LLM & Multimodal	SFT, LoRA/QLoRA, retrieval-augmented generation (RAG), prompt/context engineering
Agentic Systems	Tool use and function calling, multi-agent orchestration, model routing, MCP, agent evaluation harnesses
Inference & Efficiency	vLLM, SGLang, quantization, KV-cache optimization, ONNX, TensorRT-LLM, MLC-LLM, latency and cost profiling
Training Infrastructure	Multi-node distributed training, Slurm, Ray, Docker, Kubernetes, AWS, Azure, Git
Evaluation & Data	Benchmark design, LLM-as-judge, human evaluation protocols, large-scale data curation
Hardware & Sensing	Meta Aria Glasses, eye tracking, IMU, Raspberry Pi